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Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ both be non-empty and open and suppose $f : U \rightarrow V$ is a smooth bijection with smooth inverse (a diffeomorphism). Show that $n = m$.

Chain rule:

We have any two functions

$$f^{-1} \circ f : \mathbb{R}^n \rightarrow \mathbb{R}^n \quad (1)$$

$$f^{-1} \circ f : \mathbb{R}^m \rightarrow \mathbb{R}^m \quad (2)$$

$$f^{-1} \circ f = Id_{\mathbb{R}^n} \quad (3)$$

We recall the fact that a linear function is its own differential.

$$D(f^{-1} \circ f) = D(Id_{\mathbb{R}^n}) = Id_{\mathbb{R}^n} \quad (4)$$

Now we use chain rule on

$$D(f^{-1} \circ f) = D(f^{-1}) \circ f = Id_{\mathbb{R}^m} \quad (5)$$

Thus at any $x \in \mathbb{R}^n$, $Df_x : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is invertible so $n = m$.

HW1 Q2

Let $M = \{(t^3, t^2) \mid t \in \mathbb{R}\}$. Give a formal proof that M is not a manifold.

We will show there is no coordinate patch around $(0, 0) \in M$. Arguing by contradiction

Suppose $\beta : I \rightarrow V_{open} \subset M$ is a patch around $(0, 0)$. Suppose $\beta(0) = (0, 0)$. To check whether M can be locally parameterized as a smooth manifold, we compute its derivative:

$$\frac{d}{dt}(t^3, t^2) = (3t^2, 2t) \quad (6)$$

at $t = 0$ we get $(0, 0)$ thus $\beta(t)$ has a derivative which is zero ie $(\beta'_1, \beta'_2) = (0, 0)$. Since there is no valid local parameterization can exist around $(0, 0)$, this contradicts the assumption that M is a manifold. Thus M is not a manifold

HW1 Q3

Let $\alpha : \mathbb{R} \rightarrow \mathbb{R}^2$ be a map $\alpha(x) = (x, x^2)$ let M be the image set of α . (a) make a sketch of $M \subset \mathbb{R}^2$ (b) Show that M is a 1-manifold in $\mathbb{R}^2$ of class C^∞ , covered by a single coord patch α

Sketch is just x^2

(b) To show α is a patch for all $p \in M$ We need to check smooth, homeomorphism and $D\alpha$ rank 1 $\forall x$ α smooth? Its partials exist for all orders they are continuous Homeomorphism? We have the continuous inverse $p : M \rightarrow \mathbb{R}, (x, y) \rightarrow x \xrightarrow{\alpha} (x, x^2) \xrightarrow{p} x \xrightarrow{\alpha} (x, x^2) \xrightarrow{p} x \xrightarrow{\alpha} (x, x^2)$ $D\alpha$ has rank 1 $\forall x \in M$ as the first entry is 1

HW1 Q4

Let $U \subset \mathbb{R}^n$ be an open subset and $f : U \rightarrow \mathbb{R}^m$ be a smooth function. Show that the graph of f , that is

$$\Gamma_f := \{(x, f(x)) \in \mathbb{R}^n \times \mathbb{R}^m \mid x \in U\} \subset \mathbb{R}^{n+m} \quad (7)$$

is a smooth n -dim manifold

We propose the global chart $\alpha : U \subset \mathbb{R}^n \rightarrow \Gamma_f$. $\alpha(x) = (x, f(x))$ $f(x) = (f_1, \dots, f_m)$

$$D\alpha(x) = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ \partial_1 f_1 & \partial_2 f_1 & \cdots & \partial_n f_1 \\ \vdots & \vdots & \ddots & \vdots \\ \partial_1 f_m & \partial_2 f_m & \cdots & \partial_n f_m \end{bmatrix}$$

(8)

Manifolds with boundary Any open set $U \subseteq \mathbb{R}^n$ is a manifold
 $\forall$ manifolds are a manifold with boundary but not vice versa.

$\forall$ points either boundary or interior. Could be both but inverse function thrm says no. As a different patches would map from boundary to the interior.

The interior of a set $\overset{\circ}{A}$

Hw Qs

Problem 5 Show the following approximation result using a partition of unity. Let $A \subset \mathbb{R}^n$ be an open subset and $f : A \rightarrow \mathbb{R}^m$ be a continuous function. Show that for any $\epsilon > 0$ one can find a smooth function $g : A \rightarrow \mathbb{R}^m$ such that

$$\|f(x) - g(x)\| < \epsilon \quad \text{for all } x \in A.$$

Problem 1 Consider the subset T of $\mathbb{R}^2$ defined by

$$T = \{(x, y) \in \mathbb{R}^2 \mid y^2 = x^3 + ax + b\}$$

with $a, b \in \mathbb{R}$. a) Is T for $(a, b) = (-1, 0)$ a manifold? If not, which point(s) of T would need to be excluded to make T a manifold? b) Is T for $(a, b) = (-3, 2)$ a manifold? If not, which point(s) of T would need to be excluded to make T a manifold? c) Determine for which values of a and b T is a manifold?

Problem 2 Show that the torus

$$\mathbb{T} = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \left(R - \sqrt{x^2 + y^2} \right)^2 + z^2 = r^2 \right\}$$

is a manifold without boundary if $R > r$. (Also convince yourself that the equation indeed cuts out a torus)

Problem 3 Let $M \subset \mathbb{R}^n$ be a manifold without boundary. Let us define the "geometric" tangent space at a point $p \in M$ to be the set of vectors $v \in \mathbb{R}^n$ such that there exists a smooth curve $\gamma : \mathbb{R} \rightarrow M; t \mapsto \gamma(t)$ such that

$$\gamma(0) = p \text{ and } v = \left. \frac{d}{dt} \right|_{t=0} \gamma$$

Show that the tangent space $T_p M$ coincides with the geometric tangent space at p . Is the same true for a manifold with boundary?

Problem 4 Let $M \subset \mathbb{R}^n$ be a d -manifold with boundary. Recall that we defined the tangent bundle

$$TM = \{(p, v) \in M \times \mathbb{R}^n \mid v \in T_p M\} \subset \mathbb{R}^{n+n}$$

a) Show that TM is a manifold with boundary of dimension $2d$. b) Let M and N be manifolds with boundary and $f : M \rightarrow N$ a smooth map. Show that $Df : TM \rightarrow TN$ is also smooth. c) Let $M \xrightarrow{f} N \xrightarrow{g} L$ be smooth maps, where M, N and L are manifolds with boundary. Show that $D(g \circ f) = Dg \circ Df$.

Let V be a d -dimensional vector space. A function $T : V \times V \times \dots \times V \rightarrow \mathbb{R} = T : V^k \rightarrow \mathbb{R}$ is multi linear or a k -tensor if $\forall i, T_i(\hat{v}_i) = (v_1, \dots, v_{i-1}, \dots, v_k)$ is linear. $\mathcal{L}^k(V) = V^* = \{T : V^k \rightarrow \mathbb{R}\}$

QUESTIONS HW

0.0.1 (W1)

P1

P2

P3

P3

P4

P5

(W2)

P1

P2

P3

P4

P5

Q2 2024

(W3)

P1

P2

q 5 2024

P3

P2

P3

P4

(W4)

P1

P2

P3

P4

(W5)

P1

P2

P3

P4

P5

(W6)

P1

P2

P3

Similar to Q6 2024

P4

P5

OPTIONAL

(W7)

P1

P2

P3

P4

(W8)

P1

Q1 2024

P2

P3

(optinal)

(w9)

P1

P2

P3

DISCUSSED twice was a pp q

(w10)

P1 Talked through

P2

P3

Past q

- 1 POU
- 2 DIFF FORMS
- 3 Equivelant DEF of manifolds
- 4 Orientation
- 5 Integration
- 6 Volume forms